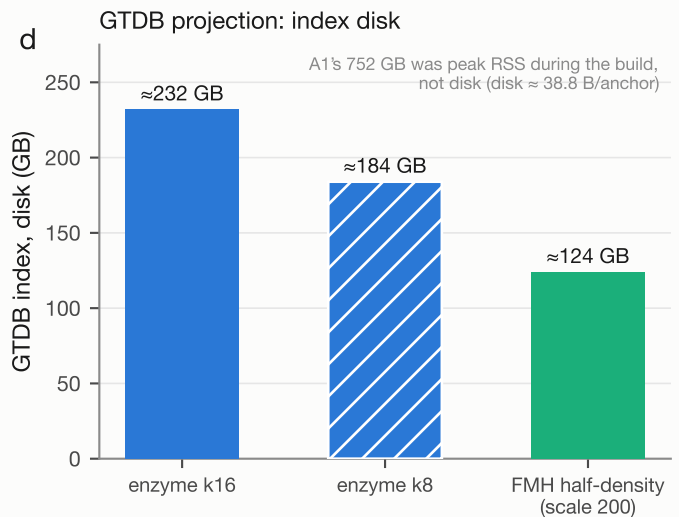
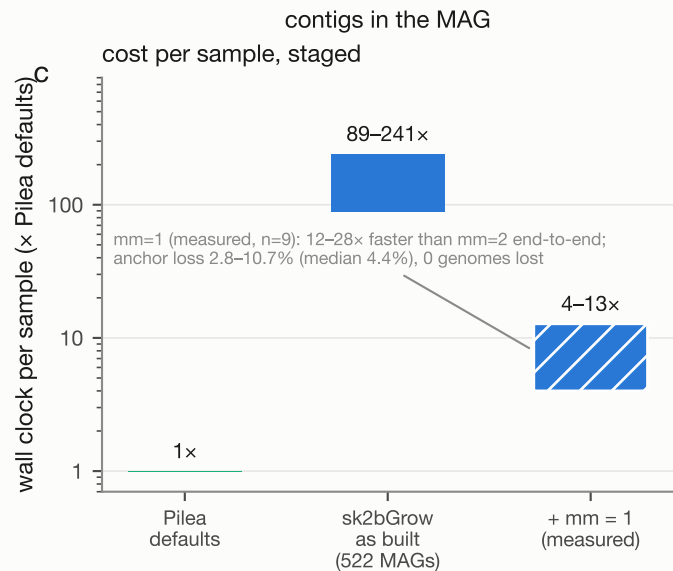
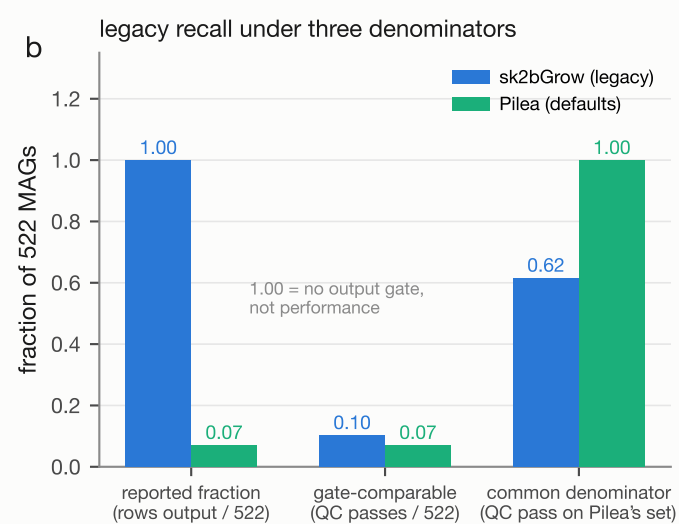
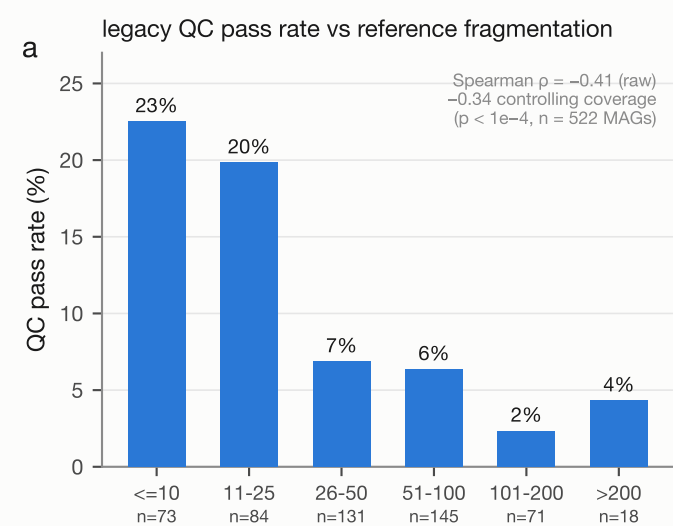




RBC metagenome (PRJNA974210): 522 MAGs × 9 samples. Panels a and b show the legacy C5 statistics, which predate the current signed fixed-origin and conservative fragmented-reference policy. a, the enzyme-consistency QC passes less often as the reference fragments: mean QC pass rate falls from 22.5% (≤ 10 contigs) to 2–7% (> 25 contigs); Spearman $\rho = -0.41$ across 522 MAGs, still -0.34 after controlling for mean coverage (both $p < 1e-4$) — legacy QC tracked fragmentation in references; coverage is the strongest single predictor and the two effects are partially entangled in this MAG population. b, legacy recall under three denominators: sk2bGrow's reported fraction is structurally 1.00 (it has no output gate) — a denominator artifact, not performance; under the gate-comparable denominator (QC passes / 522) sk2bGrow reports 3.8–13.8% and Pilea (defaults) 5.0–11.1%, overlapping ranges; on the common denominator (the MAGs Pilea reports, which sk2bGrow also estimates) sk2bGrow's QC passes 43–76% of them.

Cost narrative for scaling past single-genome benchmarks. c, wall clock per sample relative to Pilea at its defaults: on the 522-MAG RBC dataset the as-built matcher costs $\approx 90\text{--}240\times$ (matching reads against 24.1M anchors is 89.6% of count time); dropping to $--\text{max-mismatch } 1$ (two $\sim 16\text{-bp}$ seeds instead of three $\sim 11\text{-bp}$ seeds) shortens posting lists enough to cut this to $\approx 4\text{--}13\times$ measured end-to-end on all nine samples (12–28 $\times$ over the $\text{mm} = 2$ baseline), at the price of 2.8–10.7% of anchors lost (median 4.4%; 0 genomes lost; unbiased on shared anchors). At GTDB scale ($\geq 90\%$ of references absent from a sample) the M4 containment screen adds a $21.1\times$ lookup speedup and drives false positives on absent genomes from 4,048/4,700 to 0. Under the current auto policy, only 26 of 4,698 genome $\times$ sample observations pass QC because unsafe sorted-rank fallback is refused on fragmented MAGs (Table 10). d, GTDB projection (136,646 species representatives ≈ 629 Gbp, 38.8 B/anchor on disk): enzyme k16 ≈ 232 GB, k8 ≈ 184 GB, half-density FMH (scale 200) ≈ 124 GB with ≤ 0.007 r lost at $\geq 1\times$. The often-quoted 752 GB is peak resident memory during a build, not disk.